Reading tissue across magnifications

Can one research model recognize a breast-tumor subtype from the microscope views available for a case?

40× 100× 200× 400×

Main question

Choose among **eight tumor subtypes**, even when some magnifications are withheld.

Secondary question

Combine the saved subtype scores into **benign / malignant** groups.

A reproducible research prototype. Clinical use has not been established.

Many images. Few independent cases.

7,909
tissue images

4
magnifications

8
subtypes

What we group together

All related fields and magnifications stay within their provider-case role.

We do not assume that equally numbered fields align across zooms.

What remains uncertain

82 subtype-qualified provider keys;
81 bare identifiers; **70** numeric stems.

Exact clinical-patient relationships remain unverified.

More photographs of one case do not create more independent cases.

Two different experiments

SUPPLIED NOTEBOOKS · SAVED RESULTS

93.73% / 94.99%

Single-image accuracy

- One 40× image; eight classes
- Random 80/20 image split
- Validation reused for model selection
- EfficientNet attention variants

CURRENT GROUPED VALIDATION

45.7% / 38.3%

Provider-case accuracy

- Aggregate fields within each case
- All four zooms; 15 availability subsets
- Case roles enforced by the protocol
- Frozen Phikon representation

This is not a measured drop on the same test. The unit, split and model differ.

From tissue fields to eight scores

1 · Read each image

Official processor:
shortest-edge resize,
224 × 224 center crop

Frozen Phikon-v2 → 1,024 features

→

2 · Combine available views

Mean fields within each zoom
→ fitting-only normalization
→ mean available zooms

Append four availability flags

→

3 · Classify the case

Saved logistic classifier
→ eight subtype probabilities

Optional fixed benign / malignant sum

Missing views are explicit

One to four magnifications can be supplied. Every nonempty combination is evaluated.

A conventional baseline

Mean pooling is an established control. This is not a new multiview transformer.

The center crop does not encode every original pixel. Fields are pooled without invented spatial pairing.

Keep the exam separate from fitting

46 fitting cases

8 validation cases

28 excluded outer-role cases

Five overlapping provider holdouts

Two inner arrangements: **812 / 813**.
One validation case per subtype in each run.
All **15** nonempty view subsets.

The main score is macro F1

Eight classes contribute equally.
Average over subsets, then over five trials.
Accuracy and subtype recall stay visible.

22 / 20 distinct validation cases; 31 in their union. No globally pristine confirmation cohort remains.

Exact subtype recognition remains weak

ARRANGEMENT 812

0.3843

macro F1

45.7% case accuracy

ARRANGEMENT 813

0.3169

macro F1

38.3% case accuracy

Keep the frozen reference

Last-block adaptation and supervised ResNet failed their registered improvement gates.

SEPARATE MATCHED REGIONAL-VIEW EXPERIMENT

Macro F1 change: **−0.0058 / +0.0059**.
Mixed tiny effects; improvement gate failed.

Compared against its own fixed-*C* control.
The historical reference used a *C* grid;
its tuning budget is different.

Successful execution and reproducibility do not establish strong predictive performance.

The broad group is easier than the subtype

Fixed malignant score = sum of four malignant-subtype probabilities. Predict malignant at ≥ 0.5.

Saved pipeline	Accuracy	Sensitivity	Specificity	AUROC
Frozen reference	87.0 / 80.7%	98.7 / 91.3%	75.3 / 70.0%	.9617 / .9408
Adapted	86.5 / 82.0%	98.7 / 93.3%	74.3 / 70.7%	.9658 / .9558
Supervised	84.7 / 82.2%	94.3 / 87.7%	75.0 / 76.7%	.8875 / .9217

Each pair is 812 / 813; five-trial means over all 15 subsets. AUROC is ranking, not accuracy.

> **Same saved eight-class models. A different label question.**
>
> No binary fitting, calibration or threshold search. Reused validation; no binary winner selected. The eight-class gate is unchanged.

Always-malignant analytic reference: 50% accuracy (four benign / four malignant per run). The paper's 86.22% binary image result uses another protocol.

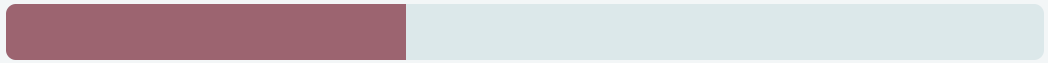
A large fitting-validation gap

Supervised ResNet

Fitting balanced accuracy **99.6-99.7%**



Validation balanced accuracy **37-40%**



Case endpoints across arrangements 812 / 813.

Very limited rare-class support

Each fitting set has just **one adenosis** case and **one phyllodes** case.

The direct image classifier is weak on validation too. Changing averaging alone did not rescue the result.

The gap is measured. No single cause has been established.

Repeated fields improve image coverage; they cannot supply new independent cases for a rare subtype.

Replay one case across available views

121
raw fields

15
view combinations

~39 s
scheduled replay run

A prespecified example

The lexicographically first validation case was chosen before inspecting its outcome.

Fresh inference reproduced all archived outputs with maximum difference **6.05×10^{-8}**.

What the viewer shows

Toggle available magnifications; inspect eight subtype scores and the fixed benign / malignant sum.

View / download replay HTML ↗

The viewer replays recorded outputs. It does not accept live image uploads.

Complete the capstone with checked evidence

Ready to present

- Audited data, grouping and source results
- Reproducible model and inference chain
- Fixed-role baselines and negative comparisons
- Independent numerical checks and demo

Still open

- Stronger transfer to exact subtypes
- Clinical identity reconciliation
- Genuinely fresh confirmation data
- A tested novel mechanism, later

November 17-18

Reviewable capstone package and revisions

→

Around November 20

Presentation · tentative
Exact date and rubric not supplied

→

After the capstone

Novel-method and paper work with explicit evidence requirements

A reproducible engineering study with useful negative findings. Clinical validation remains future work.